

WEEK 5 – TUTORIAL

Objectives:

The purpose of this tutorial worksheet is to reinforce key concepts of Project Cost Management by applying them to practical scenarios. Students will practice calculating cost and schedule performance using Earned Value Management (EVM), interpret project health through key metrics such as CV, SV, CPI, and SPI, and engage in critical thinking about the challenges and best practices of cost management in IT projects. By the end of the session, students should be able to analyse project performance data, propose corrective actions, and appreciate the importance of proactive cost control in successful IT project delivery.

Tutorial Activities:

- EVM Exercise
- Case study discussion
- Critical Thinking Questions

Learning outcomes

By the end of this tutorial, students will be able to:

- **LO2.** Apply project management knowledge, tools and techniques
- **LO3.** Demonstrate developed and refined project management skills
- **LO4.** Develop basic proficiency in project management tools and practices

Part A: EVM Exercise

Scenario:

- BAC = \$200,000, Project duration = 8 months.
- At the end of month 4:
 - Planned % completion = 50% → PV = \$100,000
 - Actual % completion = 40% → EV = \$80,000
 - AC = \$110,000

Tasks:

- Calculate CV, SV, CPI, SPI, and EAC.

$$CV = EV - AC = 80,000 - 110,000 = -\$30,000 \text{ (over budget)}$$

$$SV = EV - PV = 80,000 - 100,000 = -\$20,000 \text{ (behind schedule)}$$

$$CPI = EV/AC = 80,000/110,000 \approx 0.73 \text{ (cost-inefficient)}$$

$$SPI = EV/PV = 80,000/100,000 = 0.80 \text{ (progress slower than plan)}$$

$$EAC = BAC/CPI = 200,000/0.727... \approx \$275,000$$

$$VAC = BAC - EAC = 200,000 - 275,000 = -\$75,000$$
- What is the interpretation of the values you calculated?
Behind schedule and over budget; if trend continues, expect ~\$75k overrun.

Part B: Case study discussion

Mini Case: A university is developing an HRMS (Human Resource Management System).

- BAC = \$500,000
- Contingency Reserve = 10%

- At the end of month 6 (out of 12):
 - PV = \$250,000
 - EV = \$220,000
 - AC = \$270,000

Tasks:

- Calculate SPI and CPI.
 $CPI = 220,000 / 270,000 \approx 0.815$ (over budget)
 $SPI = 220,000 / 250,000 = 0.88$ (behind schedule)
- Interpret the results in terms of the cost and schedule.
Over budget & behind schedule.
- What might be the root cause of the performance.
Underestimated integration/testing; change requests; resource churn; vendor delays; cloud usage sprawl.
- Suggest 2 corrective actions that the project manager could take.
Scope triage & defer low-value features; resource reallocation (add senior devs to critical path); enforce change control; optimize cloud spend; negotiate vendor SLAs.

Part C: Critical Thinking Questions

- Why is cost management more challenging in IT projects compared to construction?
Intangible work, evolving requirements, rapid tech shifts, scarce talent, integration with legacy.
- How can Agile teams manage costs when requirements keep changing?
Fixed cost per sprint, backlog reprioritization, track burn rate/velocity, use “EV = budgeted value of completed stories.”
- Which best practice (from the lecture) would you prioritize in an IT project, and why?
Rigorous WBS + continuous EVM; plus cloud cost governance (budgets, alerts, tagging).